Introduction to Amazon SageMaker

Introducing SageMaker

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Course Overview

Welcome to Introduction to Amazon SageMaker

Overview

- Introduction to Machine Learning ideas
- Demonstrate SageMaker as a tool

CARVED ROCK FITNESS

Popular regional health club

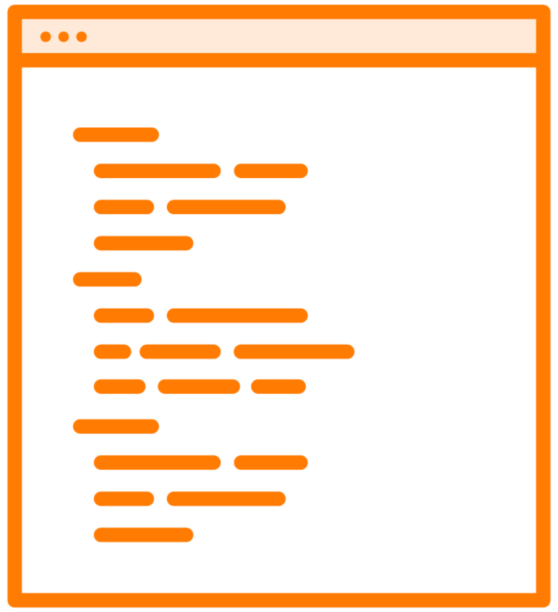
Looking to grow

Problem: Employee Absenteeism

- Large dataset of HR info
- Difficulty identifying patterns
- Limited computer resources to do analysis
- Solution: Amazon SageMaker?

Traditional Problem Solving with Computers

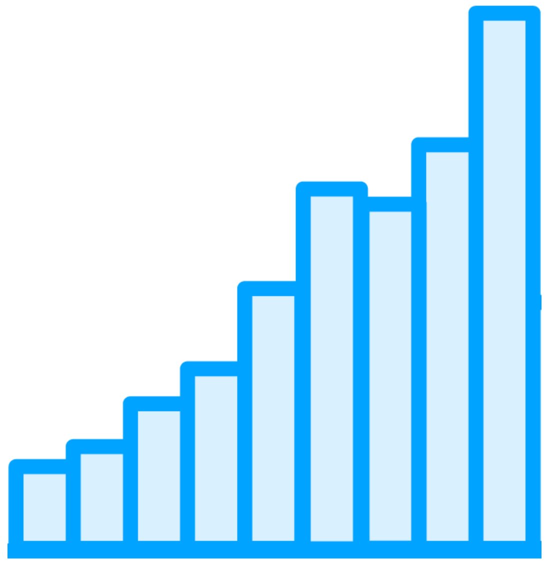
A human thinks about a problem, and develops a theory of solving it



A programmer writes software instructions to implement that solution

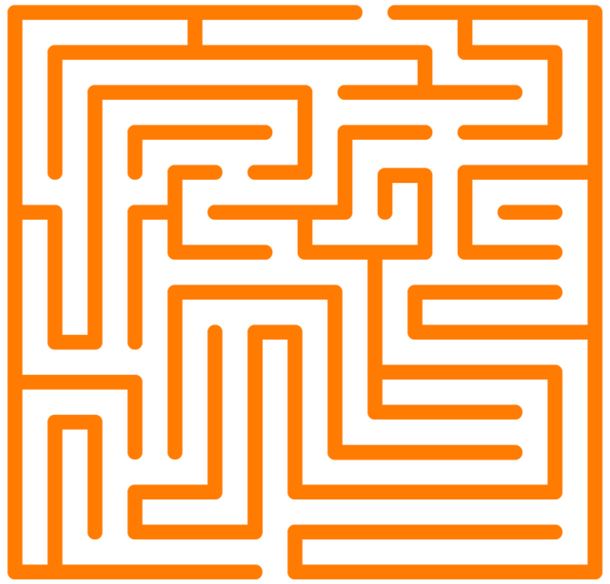
Finished software acts on inputs, taking actions or producing data

Problems for the Traditional Approach



Large Quantities of Data

Massive datasets slow down human understanding of data

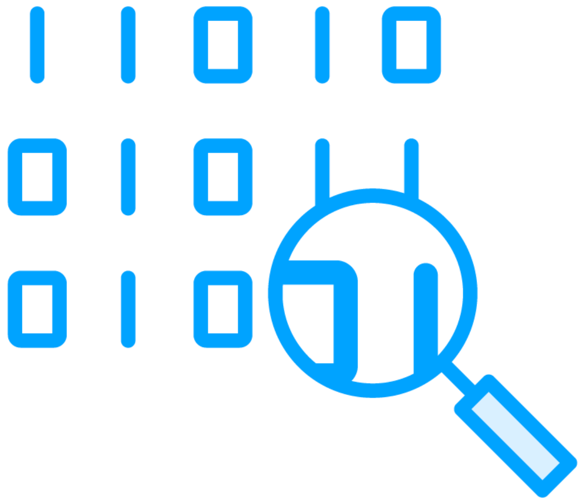


Complex Patterns

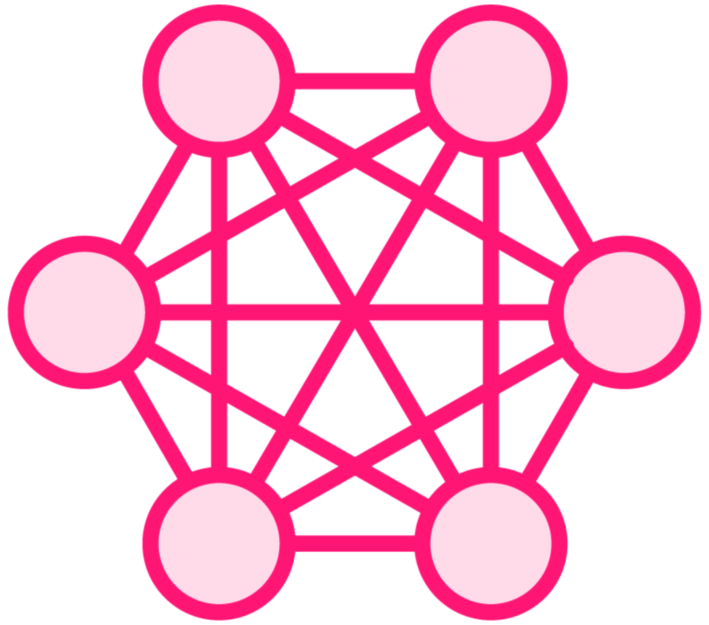
Elements of datasets can have complex, counter-intuitive interrelationships that a human might not see

Introducing Machine Learning

Computerized analysis of large quantities of data related to a problem

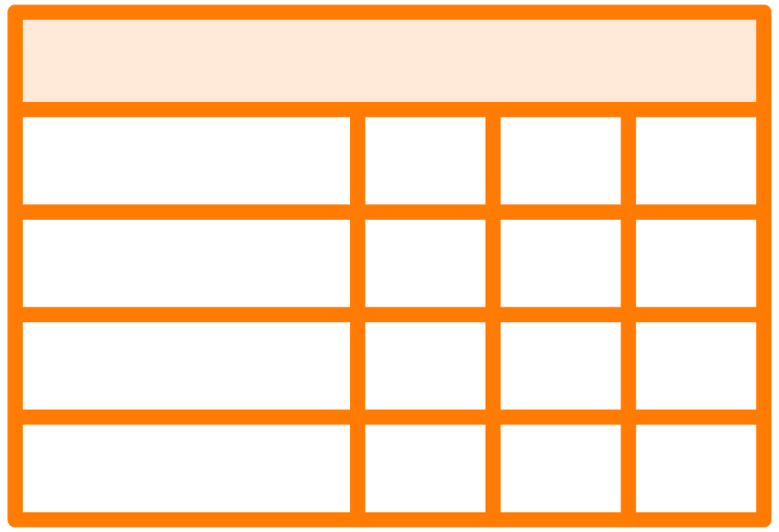


Finding statistical patterns that would be difficult or time-consuming for humans to find

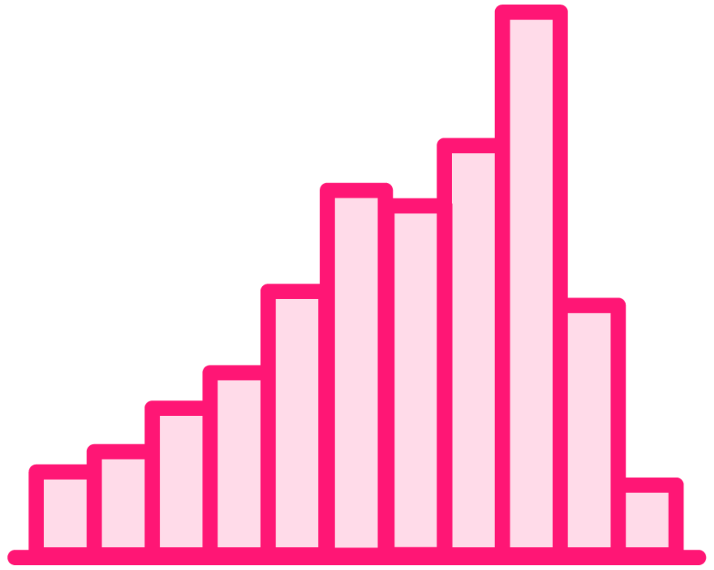


Creating a model of behavior that can be used to make predictions

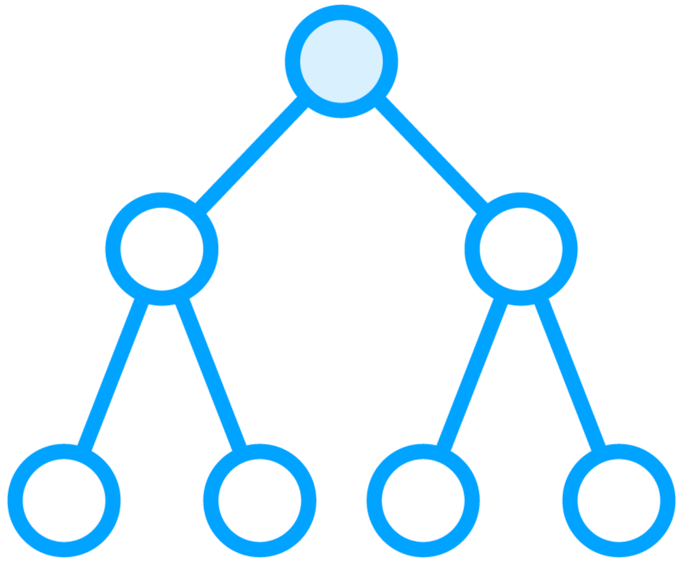
A Typical Machine Learning Project



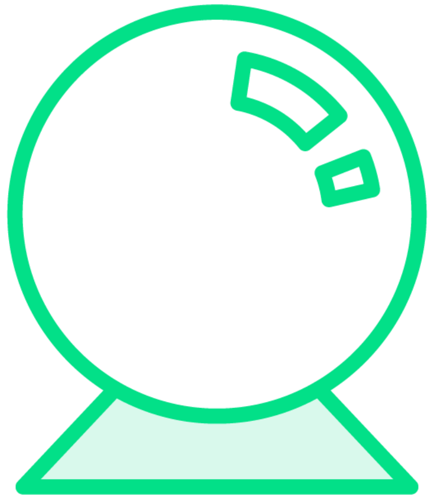
Supply a large dataset that shows inputs and output of a business process



An algorithm calculates the relationships between inputs and outputs



A model stores a mathematical description of the relationships



An application uses the model to makes predictions about new data

Problems for the Machine Learning Approach

Compute Intensive

Model creation makes heavy demands on computers, often relying on specialized GPU hardware

Data Quality

Model creation requires high-quality data. Data preparation requires specialized programming skills of Data Scientists

Solving a Machine Learning Problem with Software

Documentation	VS.	Code
Description of the data scientist's thoughts and ideas		Steps to move from a problem to a solution by manipulating data
Written in human language – often using a word processor app		Written in a programming language • Data science often uses Python
Human-readable – ignored by the computer		Computer-readable – can be hard to read for non-programmers

These are often stored and managed separately – but what if they could be combined?

Jupyter Notebooks – the Essential Tool for ML Work

Combining rich documentation and ML programming commands

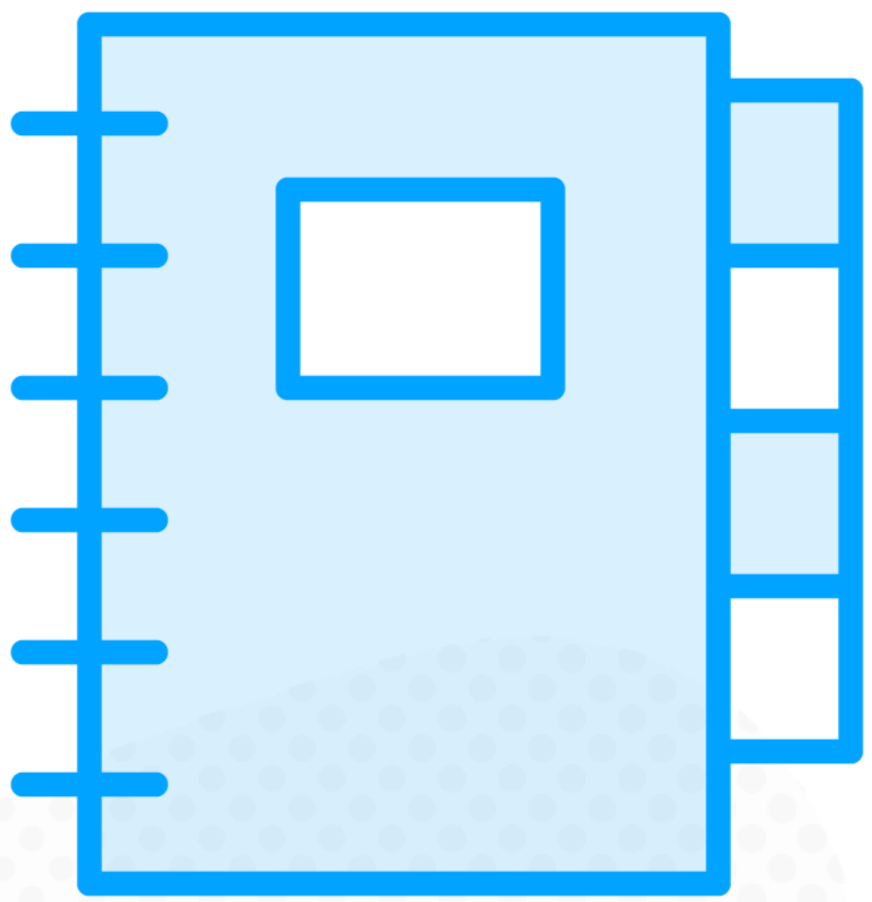
Interface feels like a web page

Developed by open-source *Project Jupyter*

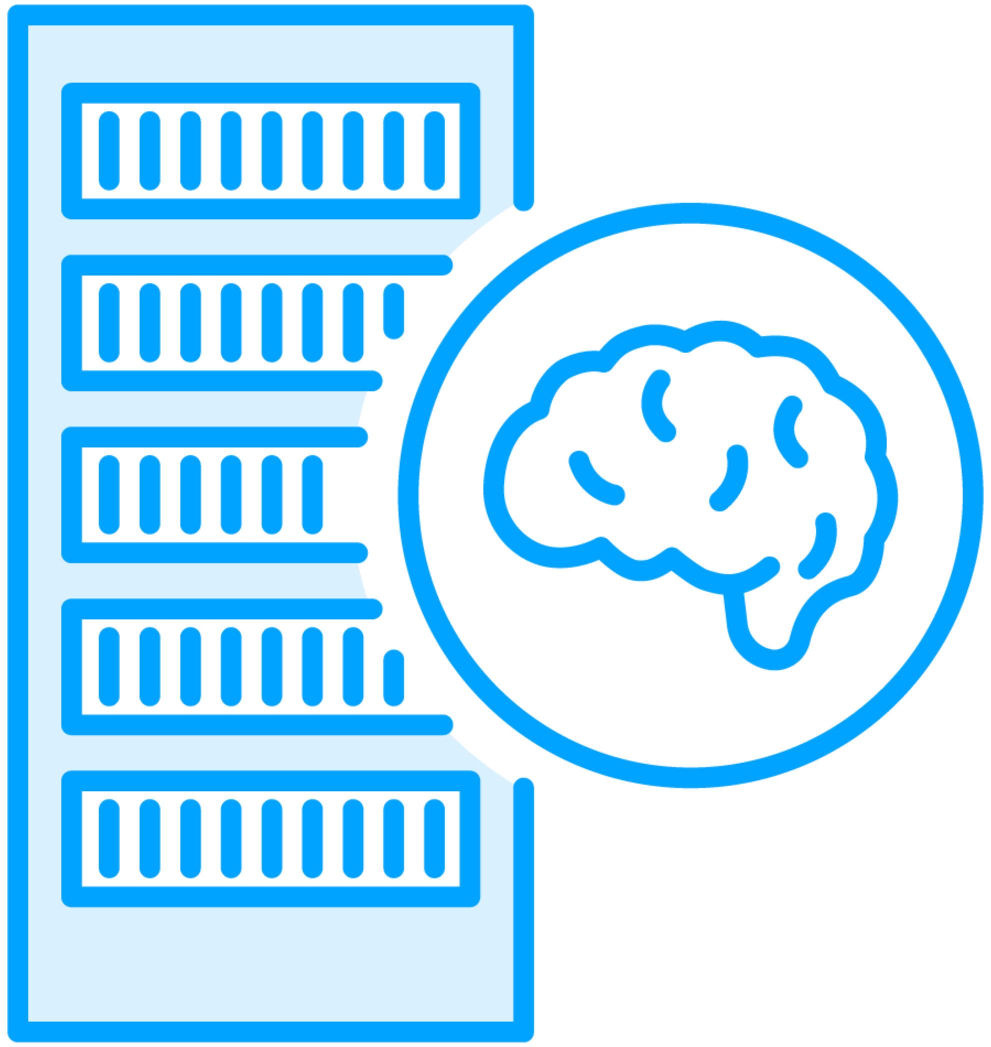
Organizes content in *cells*

Code cells – for commands

Markdown cells – for documentation



Introducing Amazon SageMaker



Cloud-based Machine Learning service

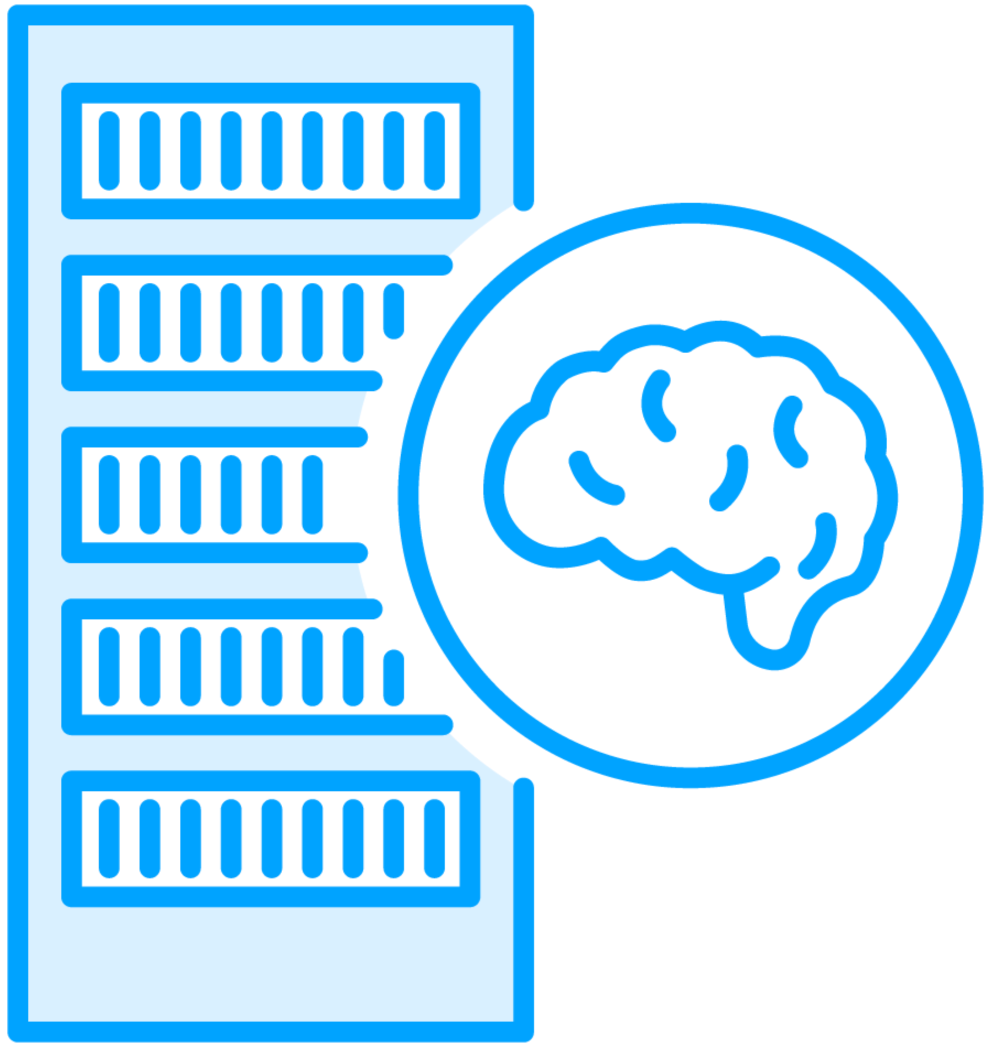
Pay only for what you use

Automated configuration of Jupyter Notebooks

Supports a full ML project lifecycle

- Data analysis and cleansing
- Model training and evaluation
- Making models available to applications
- Supporting ongoing maintenance of deployed models

SageMaker Notebook Instances



Launch a Virtual Machine in the cloud for your ML work

Automated configuration of Jupyter Notebook

Access datasets in cloud storage

Perform other ML tasks in SageMaker

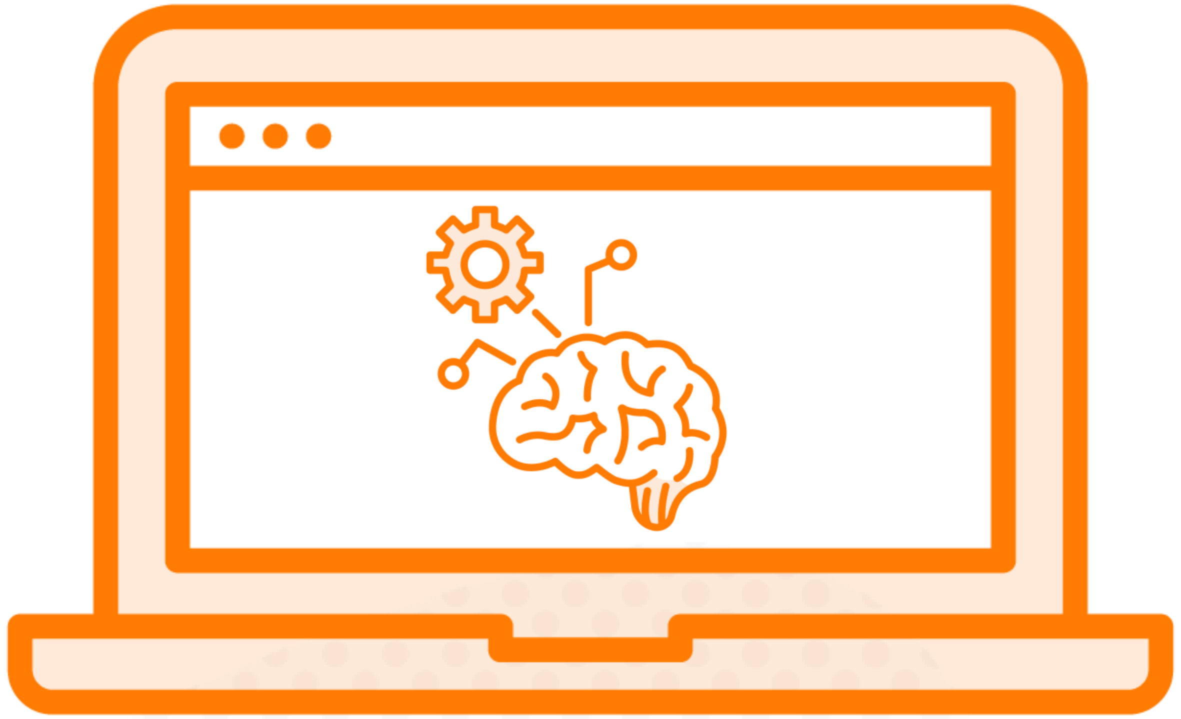
Stop the instance when you aren't working to save compute costs

The Need for a Machine Learning IDE (Integrated Development Environment)

SageMaker Instance manages storage, compute, and your Jupyter Notebook as one resource

Experience is tied to one virtual machine

New interfaces are competing with Jupyter Notebooks



Introducing SageMaker Studio

Browser-based development environment

Enables team collaboration on ML tasks

Supports a variety of development experiences
Jupyter Notebooks
Code Editor (based on VS Code)
RStudio

Separates storage and compute resources from (zero cost!) interface

Easy access to tools for Generative AI projects